

Joint use of role-playing games and models regarding negotiation processes: characterization of associations.

Abstract

Role-playing games for negotiation studies find their roots in two main families: policy exercises and experimental economics. As learning tools they aim at giving better knowledge either to players or to game organizers. They thus constitute social learning support tools: both kinds of learning are to be considered at the same time, as well as its interaction with negotiation and collective decision processes.

They are specific expressions of conceptual models, either explicitly or not. Models themselves play also role for negotiation studies, most often through computerized versions, such as Group Decision and Negotiation Support Systems.

This paper is focusing on interactions between both kinds of tools when dealing with negotiation issues. What kind of synergy may be waited for?

For that purpose, several experiments allying both tools are reviewed and compared. Many of them are coming from the thematic session on Role-playing games, models and negotiation held in Ecological Economics conference in Sousse, Tunisia, in March 2002. Two classifications of these experiments are made. One is dealing with the whole process and is leading to locate experiments according to the triangle {observed reality, controlled experiment and conceptual model}. The other is tackling more technical issues and is distinguishing according to the concomitance of use and the level of sharing conceptual model between role playing games and computerized models in joint use. This leads to consider two kinds of synergy between both tools: mutual support in use or mutual support in design.

In a post normal science context, diversity of approaches and positions lead to analyze the craft nature of these approaches and to advocate for more cautious cross-analysis of these experiments and their impacts.

Key words:

Role-playing games, negotiation, experimental science, post-normal science

Introduction

Game and play are often categorized with entertainment and fantasy. What kind of relation may exist then with research on negotiation issues, for scientific and development processes? Are they able to provide more than metaphors to the study or support of negotiation processes? However, as Huizinga pointed out in a discussion on relations between games and sacred matters, games may be something serious, and to some extent, sacred patterns and rites do share characteristics with games (Huizinga 1951). And science and development has become a new sacred paradigm, at least in developed part of the world.

Following Huizinga, let's consider that a game is a free activity, with imaginary components, without any material interest or utility, bounded in time and space, enforcing some rules and inducing group reactions (Huizinga 1951). It takes place beside "normal" social life, but is interacting with it. Even if games are supposed to take place out of normal life and without any utility or material interest for players as Huizinga is pointing out, players have their own story and experience which they cannot leave completely in the cloakroom. Caillois

distinguished four main categories of game and play: competition (*agôn*), luck (*alea*), simulation and enactment (*mimicry*) and vertigo and lure (*ilinx*) (Caillois 1967). Among those, first and third at least are at stake when dealing respectively with development or scientific experiments.

This paper is focusing on a specific category of games: role-playing games. In-between game and theater play, they are group settings determining roles or behavioral patterns of players as well as an imaginary context. A role-playing game is the performance of a sketchy situation involving people with given roles (Mucchielli 1983).

Players are genuinely using Role Playing Games as social laboratory. It is a way for them to experiment a diversity of self positioning in a group with supposedly few consequences in real world [Turtle, 1995 #236 in: (Innes 1999)]. Role playing games are also currently used as training tools and are becoming scientific tools. As group settings they seem relevant for negotiation or collective decision issues. As a training tool, they are already reported to be powerful to stimulate and support group coherent change (Tsuchiya 1998). Next step is then to envisage their suitability as group decision or negotiation support tools, while literature on Group Support Systems is more focusing on computer based instruments (Parent 2001).

These computer based instruments, mainly based on models, are supposed either through the simulation of a negotiation process or its stakes or through the building of a shared information system to support collective decision processes, raising individual and collective information. While RPG are providing participants *mimicry* through their own play, computer based models are providing it through features of its interface. Both act as "ICT tools", respectively old and new ICT tools. What kind of synergy may emerge of any joint use?

This paper is based on the analysis of communications presented at Ecological Economics conference held at Sousse, Tunisia, in 2002, within the special session "Role games, models and negotiations". This sample of experiments is completed by other papers published recently on the same topic, since such joint use of role playing games and models in collective decision setting are more and more used, even if mainly grounded in empirical approaches. All these papers analysed are dealing with the association of games and models, more specifically role-playing games and computer based models, to tackle issues of negotiation, either to get a better understanding of interaction behavioral patterns or to propose some negotiation support methodology.

Papers analysed relate several experiments enforcing such joint use of role-playing games and models for negotiation issues as well as some methodological features raised by these experiments.

This paper introduces first the origins of use of games in this field, their enforcement for negotiation issues and then their various relations to model and negotiation processes. It ends with a few practical features.

Place of games in scientific and development processes

At least two main families of games used for negotiation or dialogue issues may be found, differentiated mainly by a point of view on the process: learning tool for the players or learning tool for the game organizers. Most of them are also linked to other techniques and are increasingly considered as effective or at least potential interaction support tool.

Game as a learning tool

Games met in science or development processes have three kinds of uses: training, research or policy making (Peters 1998; Barreteau 2001). First one is however predominant. These games aim at putting players in the condition of real events in order to train them in reacting to specific conditions or to foster interactions among them according to a specific issue.

Examples of such training tools, often used with professionals in training sessions, are the most numerous. They are referenced as role playing games. Irrigation Management Game

(Burton 1989) is one of them in the field of irrigation management: players are put on a role of irrigation department officer or village water manager and have to manage an irrigated scheme. Roles are tightly defined through teacher's knowledge. However these games allow also to profit by each player own experience and real practice, such as irrigated system managers playing the irrigation management game. They are also a mean to empower trainees, decreasing the black box effect of the training support.

Another kind of game considered as a learning tool for player constitutes thought support, putting players in comparable context. The paradigm of this kind of tool might be policy exercises (Toth 1988). Players learn more on the context of their real decisions and how negotiations may evolve. With the gaming atmosphere, communication is eased since barriers among players are partly falling down (Ubbels 2000). Thus discussions among them on real cases are fostered and they may concentrate on the specific aspects chosen by the game designer. Roles are here more loosely defined while the situation is usually realistic and well documented. Experience of players is far more important. This is a very interesting alternative to learning by doing, in cases when it is hardly practically possible or ethically non acceptable, such as human systems when heavy transformations or long term planning is in discussion. It entails learning by simulating through the enforcement of the role-playing game. Such learning by simulating may also be at stake with computer based tools which in that case may have overlapping features with role playing games. This last use of game tends actually towards negotiation support through the training of negotiators on simulated cases.

Game as a tool to grasp information: when learner is on the other side

On the other side of the gaming process than players, are the observers and designers who may learn also from game sessions for their own needs. Experimental methods for social and human sciences, from economy to psychology, are absolutely in that field. These experiments are events bounded in time and space beside normal life, within a group setting (at least 2 people), regulated by very precise rules and enforce theoretical if not imaginary components. The only limit regarding Huizinga's definition of a game is that in some cases players are rewarded according to their results in the game, which is providing a specific interaction process between game world and real world. This is still consistent however with its consideration as a game, since rewards exist in many games from sports competitions to gambling. Such experimental settings are however located at one end of the range of degrees of freedom allocated to players in role playing games. Players are quite constrained, not only with rules but also through communication means. But constraints may also be fairly strict with some games with training purpose. Another feature of Role Playing Games which is barely found within these experimental settings is players having fun. But it is not compulsory to role playing games. Thus we will assimilate these experimental settings to some kind of role playing games.

Experimental economics make real players, most often but not necessarily students, put on strictly controlled roles in order to analyze their individual and collective behavioral patterns in such controlled conditions. Objectives might be: test of theories, better knowledge of human behaviors, testing of new institutional patterns (Friedman 1994). Such experiments are particularly suitable for issues dealing with the importance of communications or the influence of institutional patterns on interactions among individuals (Ostrom 1994). The controlled aspect of experiments entails possibilities of recording interactions as well as viewpoints of players in the progress of the game.

Resorting to such games positions those works in the category of experimental sciences. Between theory and observations they pave the way for a new access to knowledge (Friedman and Sunder 1994).

For that kind of use, the issue of interest of players to get in the game is important: to find players for experiments first, but also to grasp the bias due to the methodology. Therefore players are usually rewarded for their taking part in the game, with various possibilities enforcing different relations between results in the game and final reward of the player. These rewards have two main objectives according to their rationale:

- incite to participate and compensation to any direct or indirect cost of this participation. This is usually set priori to the game and is independent with the behaviour of the player during the game;
- make players interested so that they behave economically as if they were in real life.

This issue of payment of players in experimental economics is particularly at stake since it is often implicitly assumed that players want to maximize their rewards. Moreover the play aspect being weakened, why players come and play is to be analyzed thoroughly.

Another important factor affecting the dynamics of such games is the personal history of players which is not neutral: several experiments comparing results of games with heterogeneous groups of players point out the diversity of games' outcomes (Cooper 1999; Cardenas 2000; Gintis 2000). This diversity is still more important when cultures are different: Henrich and collaborators compared classic experiments with players from all over the world and got results still inconsistent with canonical theory but with more variability than experiments with students, they found out that this variability is linked to collective characteristics of players' society (Henrich 2001).

Purpose of experimentalists with this family of role-playing games is never to simulate reality, which is the main difference with the first family of games presented above. For them, game is a way to piece together a controlled complexity to work on. The emergence of this controlled complexity is coming from the multiplication of decision centers within game setting: even though each player's behaviour is strictly under control by the experimenter through fixed rules and setting, the distribution of decision processes among all players generates some complexity (Schelling 1961).

Towards interactive methods and social learning: reconsidering definition of game

Both kinds of learning are far from being exclusive. The distinction above relies mainly in the point of view on the game use, which may be both for the very same session of the very same game according to who is looking at it. With Njoobaari Ilnoowo (Barreteau, Bousquet et al. 2001), players are learning on interaction of different sets of rules while researchers are learning on other farmers' behavioral patterns (Daré 2002). This is bringing us along current trends in social sciences and decision support which are going towards social learning, with interactive methods raising the needs for specific support tools and platforms (Röling 1998).

Several role-playing games belong to such a category of interactive methods: Stratgènes game which is dealing with property rights on phylogenetic resources has been design within a close collaboration between scientists and a board of stakeholders representatives (Aubert 2002). They are actually promoting a learning through action and dialogue, which aims at being a social learning: **I learn directly as well as through your learning**. Such social learning is totally intertwined with action and practice (Friedmann 1987; Webler 1995; Daniels 1996), which is proposed to take place through role-playing games.

These games considered now as interactive methods are featuring group setting from real life for simulated stakes. Learning for players might be then only on how to interact among themselves. Intersection between game and real life is then getting bigger and bigger: game is not anymore out of normal life and it has clear utility for real activities. Relation between world of the game and world of normal life, between role playing game and social processes through "social games" appears actually to a continuum. Interaction between a society and its environment may be seen as a game in itself (Mermet 1992). Interactions within society, such as for consensus building share many points with role playing games as people act in them as representatives of groups or stakeholders and thus take on specific roles (Innes and Booher 1999).

It is not possible to separate both completely for at least role playing game is taking place within normal life. This leads to reconsider Huizinga's definition of a game: it is an activity bounded in time and space, including imaginary components, enforcing some rules, inducing group reactions and not completely embedded in normal life rather than beside it. It might not be absolutely free. It has to be an extra-ordinary meeting and be considered as a specific rendez-vous.

With social learning both game organizers and players are considered to be learning from an interactive pattern such as a role-playing game session (Hare 2002). Game organizers may be both designers and “external” observers commissioned to analyse processes at stake within the game itself. Such social learning might be a way to solve the problem of interest of players for experimental use of games, if constraints induced are suitable. While experimenters expect them to play for the rewards, they can play because they are interested in learning through the experiment.

However doing this is reaching another purpose of use of games: negotiation or discussion support, which might be already the objective of some Policy exercises. Several experiments have proved that role-playing games are always very powerful to make people discuss (Tsuchiya 1998; Barreteau, Bousquet et al. 2001; Forssén 2001) and several papers in ISEE's session relate also such features of role-playing games (Aubert, Le Page et al. 2002; d'Aquino 2002; Duijn 2002). They have been already used by their own successfully for such specific purpose with territory management issues (Piveteau 1995).

All these approaches use the ability of role-playing games to create some complexity (Schelling 1961). Even if roles cast rough but essential representations of individual behaviors, these approaches stick to the view of economy as an interactive system (Kirman 1997). It builds an artificial system through the specification and the control of some interactions among players with quite simple individual behavioral patterns. It assumes then that this designed system, notably thanks to the presence of human players, will feature other interaction patterns which are to be observed. Both controlled and observed interactions together make it a complex system. These systems are of course far more simpler than real ones, but they are featuring and simulating some complexity which is controlled and thus can be studied, as soon as suitability of scale changed is validated. They are the “like societies” on which scientists and stakeholders may think over and discuss (Kohler 1999).

By this way, games allow to reach another paradigm of “classical experimental science”: the small scale model. A role-playing game constitutes a kind of small scale model of a society in interaction with its environment. Variations among them deal with the issue to be dealt with this small scale model which will emphasize some features according to a specific question. Diversity among these approaches comes also from the enforcement of these “like societies” and their use: discussion and allocation of roles, public of players and observers, individual and/or collective debriefing.

This consideration of role-playing game as a model, leads us to analyze the relations of games with other models: computer based ones.

Relation to negotiation issues

Before analyzing the relation between games and models, since the focus is on the relations to negotiation issues, the point of view on negotiation and collective decision processes has to be specified. This is based on a post-normal position and takes in account the evolving nature of these processes.

An evolving context

Most authors agree now that collective decision and negotiation processes are continuously evolving regarding their content but also their objective and underlying values (Shakun 1996; Arthur 1997). They have an endogenous dynamics and, as open systems, evolve in relation with their context. Population of stakeholders participating in such a process evolve also with newcomers, leaving people and interested passive actors. Consensus building is considered as a “game without frontier” (Innes and Booher 1999), neither in space of solution nor in time: consensus building is steadily re-framing issues rather than finding solutions and a non-ending process, self adapting to new events (Innes and Booher 1999).

When tools involving stakeholders are used, even with a supposed minimum impact such as field observations, they generate other interactions and modify the dynamics of the process.

This evolving characteristic in interaction with tools has two consequences:

- tools must be adaptable as well or considered in a whole modeling process such as companion modeling approach (Bousquet 1999);
- influence of tools on the process dynamics must be well understood.

A post-normal position

Concerning evolving and complex systems, dealing with uncertainty, it is useless to seek for good collective decision or negotiation outcomes through powerful support tools. The legitimacy of these decisions is more important and depends more on the process which leads to it at a given moment than on the quality of the decisions themselves (Funtowicz 1999). This is post normal science and involvement of scientist in the process he or she studies or works about has to be taken in account.

It leads us to focus not only on tools, role-playing games and models, but also on methodology of use of these tools in relation with the actual world in which processes of enforcement of these methodologies take place.

Role-playing games and computerized models

Computerized models by themselves are reported to make communication around projects and plans difficult: these tools are to be considered in interaction with their users (Gardiner 1999). Moreover when dealing with society or complex systems, models need integration across disciplines as well as organizational levels. Society has a role to play in this integration of various sub-models or of various viewpoints: it can be the place for integration rather than the model itself (Robinson 1991), displaying thus a first issue of sharing competencies between computerized models and role playing games. A representation framework to describe these relations is first proposed, which serves then to describe a typology of relations between role-playing games and models dealing with negotiation.

Models, role-playing games and observed reality

It is assumed here that each type of use of role-playing game and model draws a relation between these three poles: this constitutes a proposed representation framework of experiments related and analyzed, which this paper is based upon. This means that each one might be defined by a single sentence with the three elements of the triplet {observed reality, controlled experiment, conceptual model}. For the two families of use already described, it leads to the following sentence:

Experimental economics implements **a conceptual model** in a **controlled experiment** in order to understand features of **an observed reality**.

Policy exercises use background **conceptual models** in a **controlled experiment** with stakeholders of an **observed reality**.

Each use of an association of role playing game and computerized model is linked to the three poles through the linkage sentence above. This sentence is representing the path of the process of each joint use within the triangle, stressing on their evolving characteristic. Figure 1 is featuring these paths for experimental economics and policy exercises. However most of them may be located within the triangle they constitute according to their foundations, the gravity center of their path. Figure 2 is then featuring the main location of a few examples from first part in this triangle. For example experimental economics is based on the two pillars of theoretical model and controlled experiment, while Policy Exercises are more anchored with observed reality and a little with controlled experiment, for the control and resort to model is weaker. This location is assuming that paradigm of policy exercise is a pure interactive workshop, in which no conceptual model is invited. Role playing games with training goals are closer to the controlled experiment pole because they are usually more

focused on training goal in itself rather than the reality in which training takes place. Game is providing a reality, on which action may take place (Watson 1990).

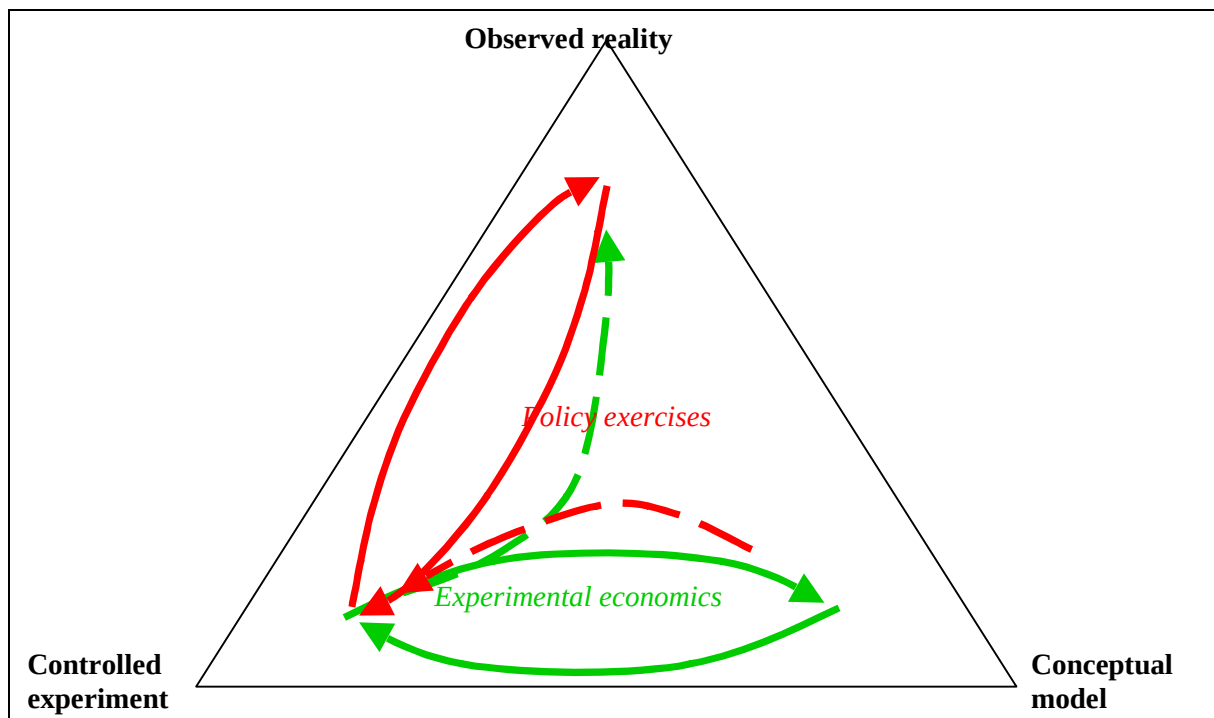


Figure 1: paths of policy exercises and experimental economics in triangle {Observed reality, controlled experiment, conceptual model}

This representation of association of role-playing games and models in such a triangle builds fully upon Legay's discussion on relations between model and experiment: model use as a scientific tool is the extension of Claude Bernard's experimental methods for complex systems (Legay 1997). The opening of the axis "Controlled experiment – Conceptual model" into a triangle is the post-normal extension of model use as a scientific tool. Models and experiments are thus still important in this point of view of science processes, but as far as social processes are concerned by the use of these tools with a post-normal posture, each experiment may be located inside the triangle proposed above. New axis, from "controlled experiment-conceptual model" edge to "observed reality" pole, is dealing with task sharing between real world, i.e. humans, and models. This sharing of tasks is dynamic and is varying along the process represented by the path in the triangle: diversity of realizations of conceptual model may be required consequently. This choice of task sharing modalities is a way of choosing the objective control rate of complexity in the experiment. It is then a matter of meeting of real world actors with this designed system as well as managing consequences of this meeting.

The third pole of the triangle is here conceptual model rather than a specific artifact of this model, usually computerized, since several artifacts of a given conceptual model might exist with a specific purpose (Barreteau 2001). One of these artifacts might be a role-playing game, actually its structure, while the pole referenced by "controlled experiment" deals with its realization with players. Model is considered as a set of roles and rules played either by a set of humans or a set of code (Hanneman 1995). Role-playing game is thus a form of model.

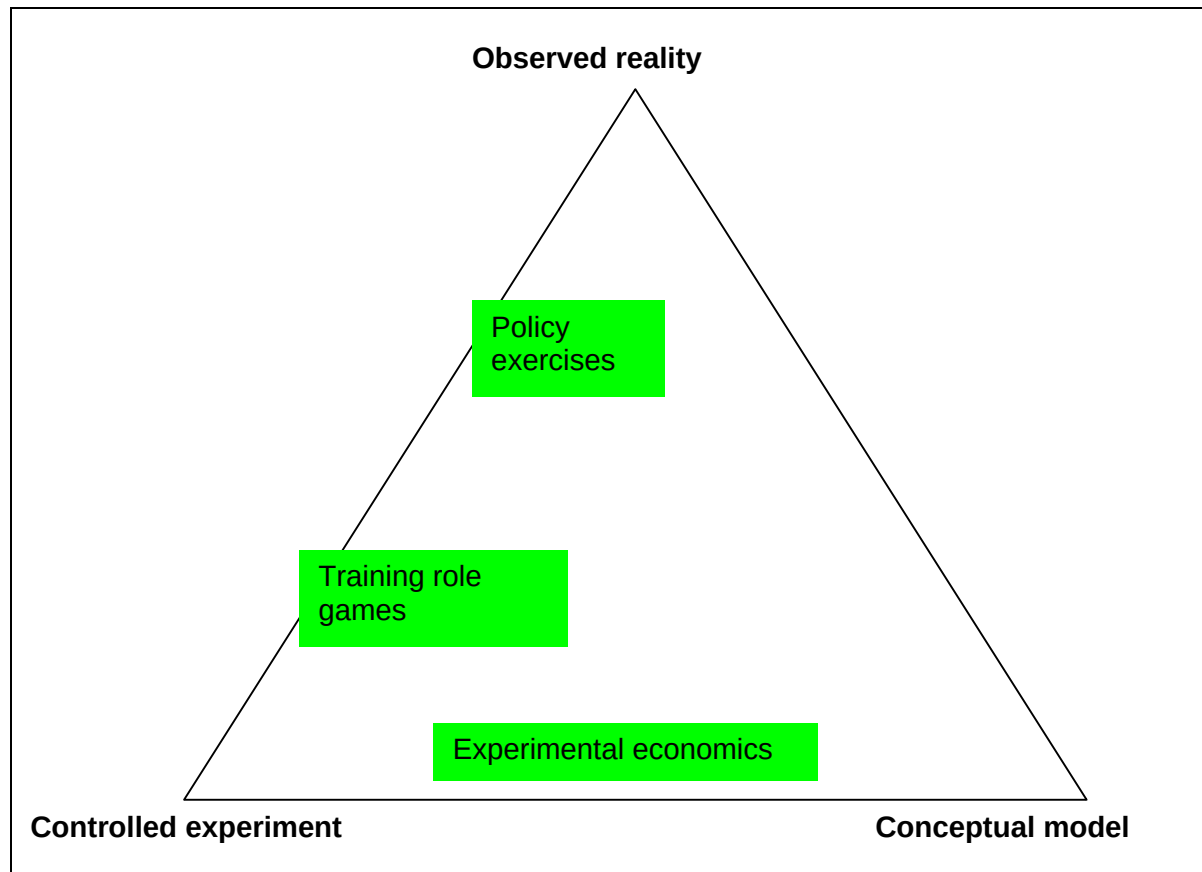


Figure 2: three poles representation of use of tools associating role-playing games and computer models

A typology of relations between role-playing games and computerized models

The diversity of associations of role-playing games and computerized models analyzed in this paper, according to their aims and their setting (public, freedom degrees allocated to players) is also found in the relation between role-playing game and computerized models. Cross analysis of this rather technical point of association of specific realization of tools in the overall processes including their use entails to deal with the issue of specific realization of task sharing modalities.

All these associations are classified with the two following keys: parts of conceptual model shared and concomitance of use. First key covers a range from two different conceptual models underlying role playing game and computerized model, while second key is dealing with the timing of use of both kinds of tools. First key should be a rather continuous criterion, but these two extreme poles are nevertheless emerging from experiments analyzed. Table 1 below is featuring these various uses according to this classification. Out of the four categories displayed by this table, more experiments fill in two of them:

- role playing game and computerized models have specific tasks which are run simultaneously and interact,
- computerized model and role playing game are used alternately according to the evolution of aim of use along the process.

The two other categories are either degenerate or rather at the border with one of the above category:

- when game and model are used to compare their results and choose the best tool, such as for prediction of outcome of conflicts (Green 2002). This use is rather on the fringe of a joint use;

- when game entails learning how to use a model (Hämäläinen 2001). This use is actually rather on the border of a concomitant use.

	different underlying conceptual models	same underlying conceptual model
model and game are used within a parallel processing	model supporting the game model included in the game game communication between model and reality	competition
model and game are used one after the other	game to learn how to use model	model of the game to repeat it game to validate model model to support game design game to support model design co-construction of model and game model as benchmark

Table 1: classification of categories of joint use of a computerized model and a role playing game according to sharing of conceptual model and relative timing of use

First major category is different underlying conceptual model and concomitant use. All the experiments falling in this category consists in a *support in use* from model to game and/or from game to model. Model might be included in the game such as when a LP model is used in a business game to show necessity of co-ordination by decision-makers through the simulation of collective decision making (Le Bars 2002). This is usually the case of business games, even though in some of them model is implicit and limited to a reproduction of reality in controlled condition and communication atmosphere (Forssén and Haho 2001). In these cases computerized models are often based on more simple conceptual models than role playing game and with a limited extension, for example to the dynamic of a resource such as in Fishbanks (Meadows 1993). Computerized model may also provide the spatial representation of consequences of choices made in game such as with simulation of consequences of road planning (Duijn, Immers et al. 2002). It goes up to a more general support of game through the use of a more or less complex model. In this category falls also, but at a large time scale of joint use, the use of game as a communication tool between a model and reality, be it to open the black box of a model (Barreteau, Bousquet et al. 2001) or to bring people to behave according to a model (Heathcote 1998). In this last case, role-playing game may also have underlying conceptual models close one another though it is not necessary useful: role-playing game aims at explaining what is in the model and may thus be a simplified version.

Second category deals with non-concomitant use and close underlying conceptual models. This means that there is only one conceptual model with various artifacts, one of them at least being a role-playing game. Computerized model is an expression of the conceptual model of the game, which might restrain modeling capacities with available computational tools, but entails a better calibration of the game and a better understanding of the model (Duffy 2001). In these cases *mutual support in design* and analysis one of the other is at stake.

Model may first entail repetitions of simulations, speeding them up and preventing by boring repetitive sessions, as Piveteau pointed out the need of computer tool with his role-playing games (Piveteau 1995). Repeats of game session with a model is not only a matter providing numerous results of simulated experiments. Computerized model allows also to repeat what happened during some specific game sessions in order to analyze them either quietly or collectively (Etienne 2002). Following this, model simulations with computer may be used as benchmarks for the analysis of game sessions. This is a computer version of the coherence

analysis driven by Etienne through a comparison of game sessions with same patterns and various players. When it is role-playing game which is used after model, it may be used for validation of a model, entailing discussion on its content by players who might know the observed reality or by making explicit the dynamics and paving the way for comparison with some real dynamics (Barreteau 1999).

In the design stage each tool may also support the other. Role-playing game is used in many experiments to elicit knowledge which becomes included in the model (Hare, Heeb et al. 2002; Le Bars, Legrusse et al. 2002). In these experiments conceptual models underlying the role-playing game are not totally formalized with some behavioral patterns not completely specified. Games entail then to improve this formalization. The whole process is heading towards making explicit implicit matters, as a matter of learning and sharing representations. Repetition and comparison of game sessions with diversity of players puts this elicitation towards more generic behavioral patterns, such as the description of a typology of behavioral patterns in negotiation of herders and foresters with repetitions of the game Sylvopast (Etienne 2002). Model may facilitate the settling of a game, through the simulation of a game prototype realization, entailing a better calibration. Games are also used to facilitate the choice of scenarios, as when stakeholders through taking on roles in game propose and fully internalize scenarios to be simulated (Hare, Heeb et al. 2002). Game may also support the design of the model, leading to discuss hypotheses in the model and thus a better social learning of model, which may even be a self-design of model by stakeholders as it has been achieved with a Senegalese rural community (d' Aquino, Le Page et al. 2002). It is one of first option with using games: exploring some patterns of interaction in order to achieve relevant theoretical models (Schelling 1961).

Overall, for many experiences related in this session, it is rather a co-construction of model and game (Etienne 2002; Hare 2002). It is a cycle involving game sessions and model simulations, each one allowing analysis and improvement of the other, leading to a permanent evolution of the conceptual model behind. Along these cycles different tools expressing the very same evolving conceptual model are used with different aims (d' Aquino 2002). This sends back to companion modeling, a three-stages cycle which may be repeated as many times as needed (Bousquet, Barreteau et al. 1999; Bousquet 2002):

- field studies and bibliography, which supply information and hypotheses for modeling and raise questions to be tackled with the model;
- modeling, i.e. converting current knowledge to a formal tool to be used as a simulator;
- simulations, done according to an experimental protocol, challenge former understanding of the system and raise new questions for a new batch of field studies...

Here, the third stage is made through a role-playing game, experiments with the only support of a computerized or a thought model being possible and fully referenced in literature.

This cycling approach is actually a path in the triangle used to locate all these possible joint uses of role-playing games and models. However when co-construction or companion modeling is concerned, all the triangle is covered and location of a gravity center representing this approach is difficult.

Figure 3 gathers all the example presented above of association of role-playing game and model used in relation with a negotiation issue. Extension of each experiment in the triangle is not meaningful: for the sake of clarity of the figure gravity center of the area covered by the path is placed. Their position relative to the three poles (Observed reality, Controlled experiment and conceptual model) may be discussed, for many of these uses may rather appear on a line from the bottom edge to Observed Reality vertex. The choice here is to fix them at the minimum level of involvement of observed reality: players' own experience or already settled co-ordination patterns. A first look at this representations shows that associations with same conceptual model for computerized model and role-playing game are more turned towards the tools and less involving observed reality: stakeholders are involved through the experimental setting constituted by the role playing game. Experiments with concomitant use are more often placed in the heart of action.

This potential extension in the triangle and its already large area covered is bringing the need for an analysis of implications of various associations of role-playing games and models for negotiation issues, be them implicit or not. This is the agenda of a beginning collective work presented in this session by (d' Aquino, Barreteau et al. 2002) on the specific case of joint use of role playing game and Agent Based Model for renewable resources management, in the following of (Bousquet, Barreteau et al. 2002). A common assumption is that players behave in the game as in reality for choices out of scope of any roles, which should be tempered since players come in game with their own habits and strategy (Daré 2002).

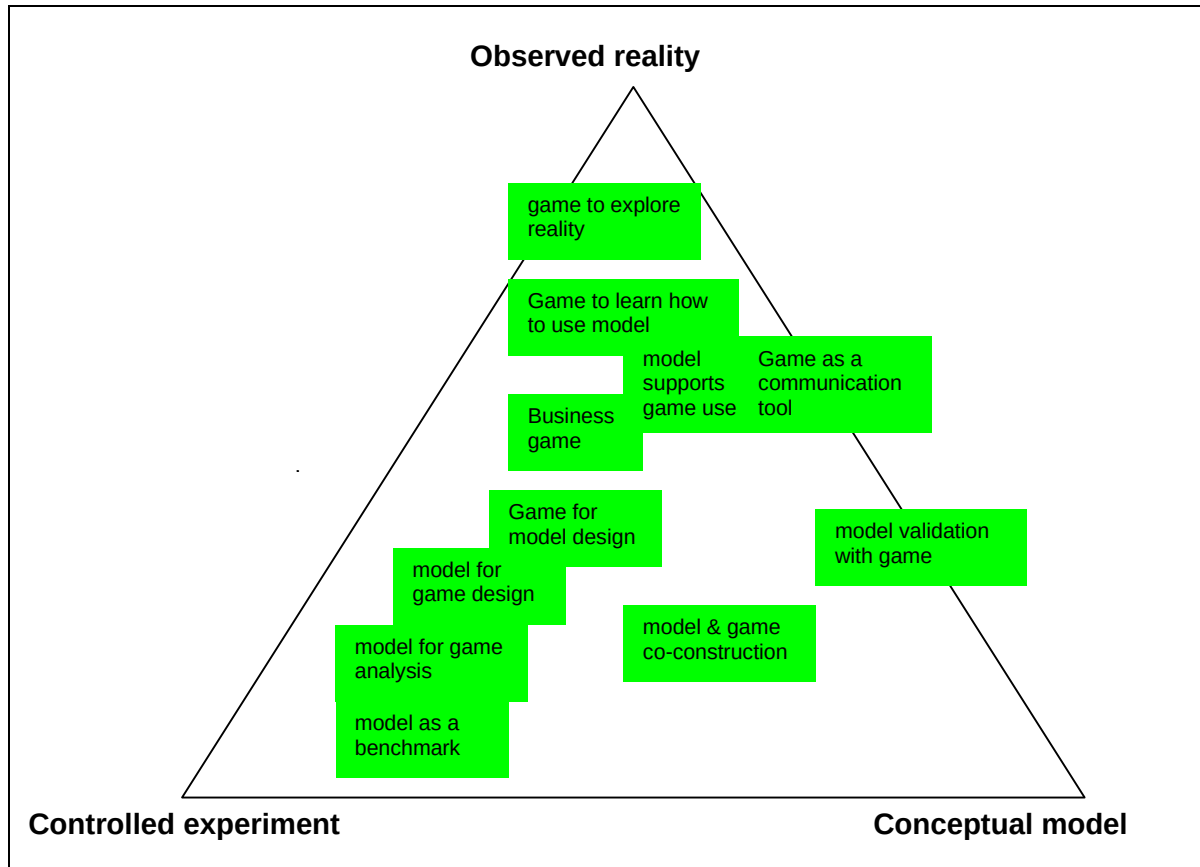


Figure 3: position of various associations in the triangle (Observed reality, controlled experiment, conceptual model).

Enforcement of such associations

Such joint use of role playing games and computerized tools for negotiation related issues is growing steadily. Most experiments are grounded empirically with field test and developments. This is in agreement with artistic nature of role playing games and pave the way for improvement of practice. However all of them agree on the learning power of this joint use which is taking a growing place in the world of artificial societies.

Artistic nature

The predominance of empirical approaches in the building of associations of games and models is coming from the acknowledged artistic nature of games (Mauriras-Bousquet 1984;Toth 1988). There are no formal strict rules to design games, but feeling and experience are mostly at stake, even if a few guidelines may be provided (d' Aquino, Barreteau et al. 2002). This artistic nature of game design is also found with scenario writing for future studies (Toth 1988). These scenarios, as the role playing games are supposed to bring players or participants in a given situation, from which they can discuss on reality

without being too much embedded in it. Artistic nature of role playing game design is not only a matter of setting design, the secret touch in the specific association of a specific number of players within a specific frame which triggers the dynamics of the game, but also a matter of conceptual model underlying and modeling choices which have been made to create a “controlled complexity”. Design of a conceptual model, be it heading towards design of a role playing game or a computerized model, is artistic.

Both elements in the associations analyzed have then artistic features. This in agreement with their focus on negotiation processes which are also reported to have some artistic or at least some craft features: most experiments analyzed are dealing with negotiation processes as some “consensus building” rather than 0-sum negotiations and Innes and Booher noticed that consensus building has to do with “bricolage” as defined by Levi-Strauss (Innes and Booher 1999). Facilitators have to do with opinions and viewpoints available in a group and reframe issues in order to build the consensus rather than reach solutions. Any end of the process is temporary and defined intuitively. Both may be considered as craftsmanship, joining then the issue of designing “long-enduring institutions” which is considered to have no best way but to be a try and error process: a crafting activity (Ostrom 1992).

Therefore, know-how in building the relation between the three poles in order to keep players interested, to get interesting outcomes without being exhausted by the first repetition is very important. The only rule which may be inferred from these experiments and analysis of various analysis (training game design, future studies, resource management, planning and political science) is to keep this relation dynamic. This place of tool and methodology of use designers know-how strengthens the necessity of ex-post analysis and comparison of experiments already mentioned.

This means that these approaches as a whole are constructivist, while each artifact, role playing game or computerized model, which can be considered as an artificial society, provides means to explore behavioral characteristics of these artificial societies and inform parties involved in dialogue process.

Practical characteristics

The enforcement of these tools and methodologies is then very pluralistic as shown with communications to this session. We are at a stage of “experimental bubbling”. Some practice related features are however already coming out.

Technical support introduces many variations: experiments go from face to face to computer based role-playing games through different sets involving cards, dices are other devices. Designers have usually tried to adapt them to their a priori opinion on what might be fine according to the actors who would be involved, the conditions in which it might take place and the objectives they had in doing such experiment. As the example with “Njoobari linoowo” game in Senegal river valley with farmers of irrigated system, “occasion cards” have been preferred to dices because of a supposed cultural reluctance to such chance control (Barreteau, Bousquet et al. 2001). The influence of such choices is still to be more analyzed.

Two kinds of such design choices are particularly interesting in presented papers for specific outcomes are awaited from them: web basis and exchange of roles.

One of the experiment analyzed is comparing face to face and web based games and points out that while face to face have easier interactions and a better control by game facilitator, web based are less costly and provides a time representation more representative of that available in decision making processes. They thus propose a co-evolution and joint use of both (Hare, Gilbert et al. 2002).

Exchange of roles (a player puts on a role near his real behavior or on the opposite the role of other kind of stakeholder’s behavior) is either advocated or left away in proposed experiments. Crossing players’ roles is interesting to increase sharing of knowledge, mutual understanding and social learning (Etienne 2002; Hare, Heeb et al. 2002). However when players keep their role they come with their own history in the actual world and use it to play (Duijn, Immers et al. 2002). This allows to elicit knowledge through the observation of players behavioral patterns. In another experiment each player’s own history is benefited from

through an exchange of roles which allows to grasp each one's point of view on the other's behavioral patterns.

Towards interacting artificial societies

Though most are still on-going works, they are already incredibly powerful tool to learn on systems we are working on. The co-evolution of model and game allows to learn on the systems both scientists and stakeholders. Role playing games and at least some of the computerized models used in these experiments, and notably agent-based models, are constituting artificial societies. Thus they provide means to understand fundamental issues which have to be brought back to real systems such as with agent-based model of Bali subaks (Lansing 2002).

But as media of interactions, as "intermediary objects" (Vinck 1999), they are also providing means to foster stakeholders' expression of viewpoints and by the way, if in an interactive setting, them to learn other viewpoints and to progress in their own knowledge of their system. The issue is then their interpretation by stakeholders and their ability to foster expression of viewpoint on real systems through the recognition of specific elements in these artificial societies which brings back to real societies, either being common to both or by remembrance.

Conclusion

Though they may appear at a first look in contradiction with the common idea of scientific and development processes, their polymorphism and a stress on their objective and methodology of use make them in association with computerized models interesting tools to understand and/or support negotiation processes as several on-going works seem to show. Several experiments of joint uses have been analyzed. They are drawing paths among three poles: observed reality, controlled experiment and conceptual model. They may be distinguished according to two factors which are the level of overlapping between conceptual models underlying role playing game and computerized model. This leads to associations more or less controlling the complexity of the systems under discussions, and to experiments which provide social learning for researchers and stakeholders as well as experiments which facilitate dialogue. Synergy of both kinds of tools come either from mutual support in use or in design. However their use is still to be analyzed for they are really in interaction with real decision processes and thus influencing it.

Anyway, their ability to generate ideas and discussions, to foster involvement of plurality of points of view and to enforce controlled complexity make them promising tools. Among their most acknowledged fruitful features are the restoring of a complexity, but a controlled one on which it is possible to work and build theories, and a shift in focus from short term to long term issues through an upset of time representation and landmarks.

Association of role-playing games with model may thus provide an extension of artificial societies (Doran 1999) through new synergic artifacts simulating complex social patterns. Their way is still to be specified and funded between an experimental sociology and a post-normal discourse on method.

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References

- Arthur W B, Durlauf S N and Lane D A, (Eds.) (1997). *The economy as an evolving complex system II*. SFI studies in the sciences of complexity. Santa Fe.
- Aubert S, Le Page C, Joly H, Razafindrabe R, Ranaivoson J F, Ralalaoherivony S B, N'Diaye I C and Babin D (2002). Conception, adaptation and diffusion of a computer-assisted role

game about phylogenetic resources management at a rural level in Madagascar. *ISEE'2002*, Sousse, Tunisia.

Barreteau O and Bousquet F (1999). Jeux de rôles et validation de systèmes multi-agents. Gleizes M-P and Marcenac P (Eds.) *Ingénierie des systèmes multi-agents, actes des 7^{èmes} JFIADSMA*: Hermès: 67-80.

Barreteau O and Bousquet F (2001). From a Conceptual Model to its Artifacts: Building on Experiments using the SHADOC Model. *MODSIM'01*, Canberra, Modelling and Simulation Society of Australia and New Zealand.

Barreteau O, Bousquet F and Attonaty J-M (2001) Role-playing games for opening the black box of multi-agent systems: method and teachings of its application to Senegal River Valley irrigated systems. *Journal of artificial societies and social simulations* 4. pp. <http://jasss.soc.surrey.ac.uk/4/2/5.html>.

Bousquet F, Barreteau O, d' Aquino P, Etienne M, Boissau S, Aubert S, Le Page C, Babin D and Castella J-C (2002). Multi-agent systems and role games : an approach for ecosystem co-management. Janssen M (Ed.) *Complexity and Ecosystem Management: The Theory and Practice of Multi-agent Approaches*: Edward Elgar Publishers: forthcoming.

Bousquet F, Barreteau O, Le Page C, Mullon C and Weber J (1999). An environmental modelling approach. The use of multi-agent simulations. Blasco F and Weill A (Eds.) *Advances in environmental and ecological modelling*: Elsevier: 113-122.

Burton M A (1989) Experiences with the irrigation management game. *Irrigation and drainage systems* 3. pp. 217-228.

Caillouis R (1967) *Les jeux et les hommes*. Gallimard.

Cardenas J-C (2000). Real wealth and experimental cooperation: evidence from field experiments. *IASCP'2000*, Bloomington.

Cooper D J, Kagel J H, Lo W and Gu Q L (1999) Gaming against managers in incentive systems: experimental results with chinese students and chinese managers. *The American Economic Review* 89. pp. 781-804.

d' Aquino P, Barreteau O, Etienne M, Boissau S, Aubert S, Bousquet F and Le Page C (2002). The role playing games in an ABM participatory modelling process: outcomes from six different experiments carried out in the last five years. *ISEE'2002*, Sousse.

d' Aquino P, Le Page C, Bousquet F and Bah A (2002). Self-Designed Role-Playing Game and Multi-Agent System to Empower a Local Decision-Making Process on Land Use Management: the SelfCormas Experiment in Senegal. *ISEE'2002*, Sousse.

Daniels S E and Walker G B (1996) Collaborative learning: improving public deliberation in ecosystem-based management. *Environmental Impact Assessment Review* 16. pp. 71-102.

Daré W (2002). Role-Playing Game In Negotiation On Irrigated Systems : Between Play And Reality. Proposal of a methodology. *ISEE'2002*, Sousse.

Doran J E (1999). Trajectories to complexity in artificial societies: rationality, belief and emotions. Kohler T and Gummerman G (Eds.) *Dynamics in human and primate societies*.: Oxford University Press: 89-105.

Duffy J (2001) Learning to speculate: experiments with artificial and real agents. *Journal of Economic Dynamics and Control* 25. pp. 295-319.

Duijn M, Immers L H, Waaldijk F A and Stoelhorst H J (2002). Gaming Approach Route 26: a combination of computer simulation, design tools and social interaction. *ISEE'2002*, Sousse.

Etienne M (2002) SYLVOPAST a multiple target role-playing game to assess negotiation processes in silvopastoral management planning. *Journal of Artificial Societies and Social Simulations* pp.

Forssén M and Haho P (2001) Participative development and training for business processes in industry: review of 88 simulation games. *International Journal of Technology management* 22. pp. 233-262.

Friedman D and Sunder S (1994) *Experimental methods, a primer for economists*. Cambridge University Press.

Friedmann J (1987) *Planning in the public domain: from knowledge to action*. Princeton: Princeton University Press.

Funtowicz S O, Martinez-Alier J, Munda G and Ravetz J R (1999). Information tools for environmental policy under conditions of complexity, European Environment Agency.

Gardiner P D and Ritchie J M (1999) Project planning in a virtual world: information management or technology going too far? *International Journal of Information Management* 19. pp. 485-494.

Gintis H (2000) Beyond homo economicus: evidence from experimental economics. *Ecological economics* 35. pp. 311-322.

Green K C (2002) Forecasting decisions in conflict situations: a comparison of game theory, role-playing and unaided judgment. *International Journal of Forecasting* forthcoming. pp.

Hämäläinen R P, Kettunen E, Ehtamo H and Marttunen M (2001) Evaluating a framework for multi-stakeholder decision support in water resources management. *Group decision and negotiation* 10. pp. 331-353.

Hanneman R A (1995) Simulation modeling and theoretical analysis in sociology. *Sociological perspectives* 38. pp. 457-462.

Hare M, Gilbert N and Pahl-Wostl C (2002). An Internet-based Role Playing Game for Developing Stakeholders' Strategies for Sustainable Urban Water Management : Experiences and Comparisons with Face-to-Face Gaming. *ISEE'2002*, Sousse.

Hare M, Heeb J and Pahl-Wostl C (2002). The Symbiotic Relationship between Role Playing Games and Model Development: A case study in participatory model building and social learning for sustainable urban water management. *ISEE'2002*, Sousse.

Heathcote I W (1998) *Integrated watershed management, principles and practice*. John Wiley & sons.

Henrich J, Boyd R, Bowles S, Camerer C, Fehr E, H. G and R. M E (2001) In Search of Homo Economicus: Behavioral Experiments in 15 Small-Scale Societies. *American Economic Review* 91. pp. 73-78.

Huizinga J (1951) *Homo Ludens*. Gallimard.

Innes J E and Booher D E (1999) Consensus building as role playing and bricolage: toward a theory of collaborative planning. *Journal of the American Planning Association* 65. pp. 9-26.

Kirman A P (1997). The economy as an interactive system. Arthur W B, Durlauf S N and Lane D A (Eds.) *The economy as an evolving complex system II*. Santa Fe: Addison Wesley: 491-531.

Kohler T A (1999). Putting social sciences together again: an introduction to the volume. Kohler T A and Gummerman G J (Eds.) *Dynamics in human and primate societies*: Santa Fe Institute: 1-18.

Lansing J S (2002). Artificial societies and the social simulations. Santa Fe, Santa Fe Institute for Complex Studies: <http://www.santafe.edu/sfi/publications/Working-Papers/02-03-011.pdf> p.

Le Bars M, Legrusse P and Attonaty J-M (2002) Three complementary and successive approaches to help collective decision-making: Budgetary Simulation, Business Game, Multi-Agent Simulation. *Journal of Artificial Societies and Social Simulations* pp.

Legay J-M (1997) *L'expérience et le modèle. Un discours sur la méthode*. INRA éditions.

Mauriras-Bousquet M (1984) *Théorie et pratique ludiques*. Economica.

Meadows D and Meadows D (1993) *Fish Banks news*. Durham, NH, USA: Fish Banks limited and Laboratory for Interactive Learning University of New Hampshire.

Mermet L (1992) *Stratégies pour la gestion de l'environnement, la nature comme jeu de société ?* L'Harmattan.

Mucchielli A (1983) *Les jeux de rôles*. Presses Universitaires de France.

Ostrom E (1992) *Crafting institutions for self governing irrigation systems*. San Francisco: ICS Press.

Ostrom E, Gardner R and Walker J (1994) *Rules, games and common-pool resources*. The University of Michigan Press.

Parent M and Gallupe R B (2001) The role of leadership in Group Support Systems failure. *Group Decision and Negotiation* 10. pp. 405-422.

Peters V, Vissers G and Heijne G (1998) The validity of games. *Simulation and gaming* 29. pp. 20-30.

- Piveteau V (1995) *Prospective et territoire : apports d'une réflexion sur le jeu*. Cemagref éditions.
- Robinson J B (1991) Modelling the interactions between human and natural systems. *International social sciences journal* 130. pp. 629-647.
- Röling N G and Jiggins J (1998). The ecological knowledge system. Röling N G and Wagemakers M A E (Eds.) *Facilitating sustainable agriculture*: Cambridge University Press: 283-311.
- Schelling T C (1961) Experimental games and bargaining theory. *World politics* 14. pp. 47-68.
- Shakun M E (1996) Modeling and supporting task-oriented group processes: purposeful complex adaptive systems and evolutionary systems design. *Group Decision and Negotiation* 5. pp. 305-317.
- Toth F L (1988) Policy exercises: objectives and design elements. *Simulation and games* 19. pp. 235-255.
- Tsuchiya S (1998) Simulation/gaming as an essential enabler of organizational change. *Simulation and gaming* 29. pp. 400-408.
- Ubbels A and Verhallen A (2000). Suitability of decision support tools for collaborative planning processes in water resources management, RIZA: 47 p.
- Vinck D (1999) Les objets intermédiaires dans les réseaux de coopération scientifique. *Revue Française de Sociologie* 40. pp. 385-414.
- Watson D R and Sharrock W W (1990). Realities in simulation/gaming. Crookall D and Oxford R (Eds.) *Simulations, gaming and language learning*. New York: Newbury House Publishers: 231-238.
- Webler T, Kastenholz H and Renn O (1995) Public participation in Impact Assessment: a social learning perspective. *Environmental Impact Assessment Review* 15. pp. 443-463.